

Advanced Microeconomics

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Problem Set 4: Applications of income and substitution effects

1) The utility function of an individual is $U(x_1, x_2) = \ln x_1 + \ln x_2$. The prices in period 0 are $p_1^0 = 1$ and $p_2^0 = 4$, the income is $w = 200$.

- a) Calculate the Marshallian demand functions, the indirect utility function, the expenditure function and the Hicksian demand functions.
- b) Suppose the government introduces a tax t on good x_1 so that $p_1^1 = 2$. Calculate the Compensating Variation and the Equivalent Variation (EV) and interpret your results.
- c) Calculate the “excess burden of taxation” associated with the new tax on good x_1 as $|EV| - t \times h(p^1, U^1)$, where $|EV|$ is the absolute value of the Equivalent Variation from (b), t is the tax, $h()$ the Hicksian demand, p^1 the new price level and U^1 the utility with the new price level. Why does this excess burden of taxation arise?

2) The utility function of an individual is given by $u(x_1, x_2) = x_1^{\frac{1}{4}} x_2^{\frac{3}{4}}$. The wealth of the individual and the prices of good 1 and 2 are greater than 0 ($w, p_1, p_2 > 0$).

- a) Determine the Marshallian demand functions.
- b) Determine the indirect utility function.
- c) Determine the expenditure function and the Hicksian demand functions.